Class Notes

DDD 2: Introduction to Programming and Basic Data Analysis

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My class notes in R. The code is shown as I wrote it in class; it is not run when the site is built.

Setup

```
rm(list=ls())


library()
```

Clears the workspace and sets the working directory.

Matrix, factor and list

```
m1<- matrix(c(0,0,1,0,0,0,0,1,1,0,0,0),nrow=4,byrow=TRUE)
m1

v1<- c("A","E","C","D")
f1<- factor(v1, levels=c("A","C","D","E"), ordered = T)
f1

f2<- factor(c("a", "b", "c", "a", "b", "c")) # categorical variableal
f2

n1<- as.numeric(f2)
n1

l1<- list(
  1:3,
  a=c("a","b"),
  c(T,F,T,F),
  c(2,3,5.6))
typeof(l1)
str(l1)
```

Builds a matrix, an ordered factor and a list, and checks their types.

Tibble and data frame

```
# difference between tibble and data.frame
# we always need r to use tibble package
library(tibble) # used to create data frame
df1<- tibble(x=1:3, y=letters[1:3]) # we dont need to check indicators of data
df1
df2<- data.frame(x=1:3, y=letters[1:3])
df2
df3<- tibble(x=1:3,y=x*2)
df3

?tibble

# generating rows names in data frame
df4<- data.frame(
  age= c(25,30,35),
  gender= c("M","F","M"),
```

```
  row.names = c("ID1","ID2","ID3"))
df4
```

A tibble does not change names or types and lets a new column use an earlier one; a data frame can carry row names.

Missing values and z-scores

```
library(dplyr)

df <- data.frame(
  a = c(1, 2, 3, 4),
  b = c(6, 8, NA, 7),
  c = c(5, 7, 8, 9))


f<- function(x){x=(x-mean(x)/sd(x))}
bs<- f(df$b)
bs




df <- df %>%
  mutate(
    c = ifelse(c == 8, NA, c), #ifelse(test, yes, no)
    b = ifelse(is.na(b), 2, b))
df

# ifelse commend used to remove outliers , like replace -ve income by na , this
is data cleaning


z_score <- function(x) {
  (x - mean(x, na.rm = TRUE)) / sd(x, na.rm = TRUE)}

df |>
  mutate(
    across(c(a, b, c, d), z_score))

# Apply z-score function to columns a, b, c, and d.
```

```r
z_score <- function(x) {
  (x - mean(x)) / sd(x)
}

df |>
  mutate(
    across(c(a, b, c, d), z_score)
  )
```

`ifelse()` replaces unwanted values with `NA` (data cleaning), and `across()` applies one function to several columns.

Environments

```r
environment()
ls()
rm(list= ls())
x<-50

# save same variable in different environments
new_envv<- new.env()
new_envv$x<- 50
ls(new_envv)
parent.env(new_envv)
#rlang used to create a new env with many variables
library(rlang)
e2<- env(a= 120, b= 100, z=20)

# function creates it is own env
f<- function(){a<-4
b<-4
a+b
}
f()
```

Variables live in environments; a function creates its own environment.

For loops

```r
# LOOPS IN R
# to run same task many times repeating in same form
loops<- new.env()
```

```r
loops$x<- c(1,2,3,4,5)

for (i in loops$x) {
  print(i^2)
}

loops$y<- 1:20
for (i in loops$y) {
  print(i^2)
}


loops$z<-c(20,30,40)
for(r in loops$z){print(r^2)}

for(a in loops$a<-c(1,2,3,4,5)){print(a^2)}

loops$xx<- numeric(5)
for(w in 1:3){loops$xx[w]<-w^2}


loops$e<- c(2,6,9)
loops$f <- numeric(length(loops$e))
for(f in 1:length(loops$e)){loops$f[f] <- loops$e[f]^2}
loops$f


#
x <- 1:5
for (i in x) {
  print(paste0(i, i^2,  i^3))}




for (i in 1:5) {
  for (j in 1:10) {
    print(i * j)
  }
}
```

```r
x <- c(4, 7, 10, 13, 16)
z_values <- numeric(length(x))
for (i in 1:length(x)) {
  z_values[i] <- (x[i] - mean(x)) / sd(x)}
z_values



loops$table <- 2:4

for (i in loops$table) {
  cat("\nTable of", i, "\n")
  for (j in loops$table) {
    cat(i, "*", j, "=", i*j, "\n")
  }
}




# random variables :- the outcomes of the an experiment
x<- c(-1,3,4)
for (i in x){if (i<0) {print("Negative")} else
  {print("Positive")}}


## nested for loop
for (i in vec1) {
  for (j in vec2) {
    for (k in vecn) {

      # your code here

    }
  }
}



for (i in 1:3){
  for(j in 1:2){
    print(i+j)
  }
```

```
}
```

```
f<- function(x){
  x^2
}

# save these results of the functions
v <- c(1,2,3,4,5,6)
x <- numeric(length(v))  # it generated empty vector  holding empty values
for (i in 1:length(x)) {
  x[i] <- (v[i])^2
}
x


for (i in 1:3) {
  for (j in 1:3) {
    print(c(i, j))
  }
}
```

A for loop repeats the same task; the results are saved in a vector created before the loop.

While and repeat loops

```
# while loop

#while(logical condition) {command here}

x<- 1
while(x<=5){print(x)
  x<-x+1
}

# position of the code matters alot
while(x<=5){  x<-x+1
```

```
print(x)
}


# store the  result
x<-1
y<- c()
while(x<5){y<- c(y,x)
  x<- x+1
}

#y<-c(y,x) “Take the current values in y, add x to them, and save the result
again in y.”

x<-1
y<- c()
while(x<5){y<- c(y,x)
x<- x+1
}

x<- 1
while(x<14){print(x)
  x<-x+2
}


# repeat loop
# infinite loop
x<-10
repeat{x<-x+1
  print(x)
}

# to break the loop
x<-10
repeat{x<-x+1
  print(x)
  if(x>15){break}
}
```

`while` runs while a condition holds; `repeat` runs until `break`.

The map family

```
# the map function
# we use map function to apply a function to each element of a list or vector
and return a list of the results.
library(purrr)
map(x,f) # x is data, f is function, it will apply function to each element of
x and return a list of the results
map() # makes a list
map_lgl() # return logical vector
map_dbl() # return double vector
map_chr() # return character vector
map_int() # return integer vector

f1<- function(x){x^3}
# first line
map(1:3, f1) # stored in data frame
map_dbl(1:3, f1) # stored in vector and values are in vector formats
map_dbl(1:3,function(x){x^3})
map_dbl(1:3, ~ .x^3)
#~.x is used to avoid writing  function(x) every time.



# two apply maps for two variables
pmap_dbl(list(1:3, 4:6), function(x,y){x^y})


# for dataframe
 df<- tibble:: tibble(rnorm(5), b=rnorm(5), c=rnorm(5))
```

`map()` applies a function to every element; the suffix (`_dbl`, `_chr`, ...) sets the output type.

Data handling and cleaning

```
 # data handling and EDA (clean, and use data for analysis)

 # Categorical (factor command ) - 1. nominal ( order does not matter ) 2,
ordinal ( order matters))
 # continuous (numerical) - 1. discrete (countable) 2. continuous (uncountable)
 # binary (logical) - 1. TRUE 2. FALSE
 # indicator variable from a categorical variable ( 0 and 1)
```

```
# 1. SET WORKING DIRECTORY

# 2. LOAD PACKAGES
 library(rio) # used to import data in R
#3. IMPORT DATA
hh_data <- import("../data/ddd2/RAW_DATA/ICPSR_36151/HH_DATA/HH_DATA.dta")
#4 SubSET DATA
hh_data|> select(ID11,ID13,FM1,ID18C) |> export("../data/ddd2/AUXILIARY_DATA/
hh_data_final.csv")
```

```
Error in `select()`:
! could not find function "select"
```

```
hh<- import("../data/ddd2/AUXILIARY_DATA/hh_data_final.csv")
```

```
Error:
! No such file: ../data/ddd2/AUXILIARY_DATA/hh_data_final.csv
```

```
# to remove old raw files and keep only the final data
rm(hh_data)


attach(hh) # attach command is used to access the variables in the data frame
without using the $ sign
```

```
Error:
! object 'hh' not found
```

```
library(labelled)
var_label(hh$ID13)<- "Caste Category" # we can also use var_label(hh$ID13) to
check the label of the variable and give labels to variables
```

```
Error:
! object 'hh' not found
```

```
var_label(hh$ID11)<- "Region of  household head"
```

```
Error:
! object 'hh' not found
```

```
# if wee will work collectively on two variables, we need to delete the data which holds missing values but, if we are dealing with single variables, we just need to replace the missing with na .
library(dplyr)
hh<-hh|> rename(land= FM1, education=ID18C, caste_category=ID13, region=ID11)
```

```
Error:
! object 'hh' not found
```

```
var_label(hh$land)<- "Land"
```

```
Error:
! object 'hh' not found
```

```
var_label(hh$education)<- "Education"
```

```
Error:
! object 'hh' not found
```

```
# clean the data for variables
# first attach()
attach(hh)
```

```
Error:
! object 'hh' not found
```

```
unique(education) # to check the unique values in the variable educations
```

```
Error:
! object 'education' not found
```

```
table(education) # to check the frequency of each unique value in the variable education
```

```
Error:
! object 'education' not found
```

```
attributes(caste_category) # to check the labels of the values inside variable education
```

```
Error:
! object 'caste_category' not found
```

```
table(caste_category)
```

```
Error:
! object 'caste_category' not found
```

```
hh|>count(caste_category) # to check the frequency of each unique value in the variable
```

```
Error:
! object 'hh' not found
```

```
# how to check outliers in the datars
hh<- hh|> mutate(education= ifelse(education==16,NA, education)) # to replace the outliers with NA
```

```
Error:
! object 'hh' not found
```

```
table(hh$education)
```

```
Error:
! object 'hh' not found
```

```
summary(hh$education)
```

```
Error:
! object 'hh' not found
```

```
# creatting new education variables ( creating dummy for continuous variable)
hh<- hh|>
  mutate(any_edu= ifelse(education>0,1,0| is.na(education))) # to create a dummy variable for education, if education is greater than 0, then any_edu is 1, otherwise it is 0. We also need to check for NA values in education variable and assign 0 to any_edu for those cases.))
```

```
Error:
! object 'hh' not found
```

```
var_label(hh$any_edu)<- "Any education"
```

```
Error:
! object 'hh' not found
```

```
# categories of education (1.2.3.....)
hh<- hh|>
  mutate(edu_cat= case_when(
    education<= 5 ~ 1,
    education > 5 & education <= 12 ~ 2,
    education > 12 ~ 3,
  ))
```

```
Error:
! object 'hh' not found
```

```
var_label(hh$edu_cat)<- "Education categories"
```

```
Error:
! object 'hh' not found
```

```
val_labels(hh$edu_cat)<- c(primary=1, secondary=2, higher=3)
```

```
Error:
! object 'hh' not found
```

```
attributes(hh$edu_cat)
```

```
Error:
! object 'hh' not found
```

```
table(hh$edu_cat)
```

```
Error:
! object 'hh' not found
```

```
hh<- hh|> filter(education<=15&education>0| is.na(education))
```

```
Error:
! object 'hh' not found
```

Imports the IHDS household data, keeps four variables, adds labels, replaces outliers with `NA` and builds education categories.

Summary statistics

```
# summary statistics
HH_research |>
  dplyr::select(total_income, education_max) |>
  stargazer(
    type = "text",
    title = "Summary Statistics",
    summary.stat = c("n", "mean", "sd", "min", "max"),
    digits = 2)


# This gives you ONE table with caste as columns
HH_research |>
  dplyr::filter(!is.na(caste_cat)) |>
  dplyr::mutate(
    total_income = as.numeric(as.character(total_income)),
    education_max = as.numeric(as.character(education_max))
  ) |>
  group_by(caste_cat) |>
  summarise(
    N = n(),
    Mean_Income = mean(total_income, na.rm = TRUE),
    SD_Income = sd(total_income, na.rm = TRUE),
    Min_Income = min(total_income, na.rm = TRUE),
```

```
    Max_Income = max(total_income, na.rm = TRUE),
    Mean_Edu = mean(education_max, na.rm = TRUE),
    SD_Edu = sd(education_max, na.rm = TRUE))
```

Summary table for income and education, and the same by caste category.

Bar plots

```
# data visualization
library(rio)
library(dplyr)
library(labelled)
library(ggplot2)
library(stargazer)

# set working

# BAR PLOT simple variable  here x= consider as numerical variable
ggplot(hh, aes(x= edu_cat)) +
  geom_bar() # NAs are not included
```

```
Error:
! object 'hh' not found
```

```
#BAR PLOT by using factor to say that the variable is categorical
ggplot(hh, aes(x= factor(edu_cat))) +
  geom_bar() # NAs are included
```

```
Error:
! object 'hh' not found
```

```
ggplot(hh, aes(x= factor(edu_cat), fill= factor(edu_cat))) +
  geom_bar() # add legends and colour
```

```
Error:
! object 'hh' not found
```

```
# like here NA is included in our graphs , and, i dont need na so usie filter
to remove NA values
hh|> filter(!is.na(edu_cat)) |>
```

```
  ggplot(aes(x= factor(edu_cat), fill= factor(edu_cat))) +
  geom_bar() # now NA is not included in our graph
```

```
Error:
! object 'hh' not found
```

```
ggplot(hh|>    filter(!is.na(edu_cat)),    aes(x=    factor(edu_cat),    fill=
factor(edu_cat))) +
  geom_bar() # same as above but in one line code
```

```
Error:
! object 'hh' not found
```

```
# how to remove legends
#way1
ggplot(hh|>    filter(!is.na(edu_cat)),    aes(x=    factor(edu_cat),    fill=
factor(edu_cat))) +
  geom_bar() +
  theme(legend.position = "none") # to remove legends
```

```
Error:
! object 'hh' not found
```

```
# way2
hh|> filter(!is.na(edu_cat)) |>
    ggplot(aes(x=  factor(edu_cat,  levels=  c(1,2,3),  labels  =  c("Primary",
"Secondary", "Higher")), fill= factor(edu_cat))) +
  geom_bar()+
  theme(legend.position = "none")
```

```
Error:
! object 'hh' not found
```

```
# best code for plotting the graph with all the details
hh|> filter(!is.na(edu_cat)) |>
    ggplot(aes(x=  factor(edu_cat,  levels=  c(1,2,3),  labels  =  c("Primary",
"Secondary", "Higher")), fill= factor(edu_cat))) +
  geom_bar()+labs(
```

```
    x = "Education Categories",
    y = "Count",
    title = "Distribution of Education Categories",
    fill = "Education Categories"
  )+
  theme_minimal()+# to change the background of the graph
  theme(
    plot.title = element_text(hjust = 0.5))+ # to center the title
  theme(legend.position = "left")
```

```
Error:
! object 'hh' not found
```

```
# but this is not useful, we need to add proportions in the graph, so we need
to use geom_text to add the proportions in the graph
hh|> filter(!is.na(edu_cat)) |>
  count(edu_cat) |> mutate(prop = n / sum(n)) |>
    ggplot(aes(x=  factor(edu_cat,  levels=  c(1,2,3),  labels  =  c("Primary",
"Secondary", "Higher")), y = prop,  fill= factor(edu_cat))) +
  geom_col()+labs(
    x = "Education Categories",
    y = "Proportion",
    title = "Distribution of Education Categories",
    fill = "Education Categories"
  )+
  theme_minimal()+# to change the background of the graph
  theme(
    plot.title = element_text(hjust = 0.5))+ # to center the title
  theme(legend.position = "left")
```

```
Error:
! object 'hh' not found
```

```
library(scales)

# converting y axis in percentage
#   add   after   geom_col()   to   convert   y   axis   in   percentage,   use
scale_y_continuous(labels= percent) to convert y axis in percentage
hh|> filter(!is.na(edu_cat)) |>
  count(edu_cat) |> mutate(prop = n / sum(n)) |>
    ggplot(aes(x=  factor(edu_cat,  levels=  c(1,2,3),  labels  =  c("Primary",
```

```
"Secondary", "Higher")), y = prop,  fill= factor(edu_cat))) +
  geom_col()+ scale_y_continuous(labels= percent)+labs( ### percentage
    x = "Education Categories",
    y = "Percentage",
    title = "Distribution of Education Categories",
    fill = "Education Categories"
  )+
  theme_minimal()+# to change the background of the graph
  theme(
    plot.title = element_text(hjust = 0.5))+ # to center the title
  theme(legend.position = "left")
```

```
Error:
! object 'hh' not found
```

```
# add percentage at the top of the bars
hh|> filter(!is.na(edu_cat)) |>
  count(edu_cat) |> mutate(prop = n / sum(n), top_lab= paste0(round(prop*100,1),
"%")) |> # creating new variable top_lab to add percentage at the top of the bars
    ggplot(aes(x=  factor(edu_cat,  levels=  c(1,2,3),  labels  =  c("Primary",
"Secondary", "Higher")), y = prop,  fill= factor(edu_cat))) +
  geom_col()+ scale_y_continuous(labels= percent)+
  geom_text(aes(label = top_lab), vjust = -0.5, size= 2) + # add geom_text to
add percentage at the top of the bars, vjust is used to adjust the position of
the text, size is used to adjust the size of the text
  labs( ### percentage
    x = "Education Categories",
    y = "Percentage",
    title = "Distribution of Education Categories",
    fill = "Education Categories"
  )+
  theme_minimal()+# to change the background of the graph
  theme(
    plot.title = element_text(hjust = 0.5))+ # to center the title
  theme(legend.position = "left")
```

```
Error:
! object 'hh' not found
```

```
######3 add side by side bars
#1. in filter we need to another variable
```

```
#2. in aes we need to add fill= factor(land) to add side by side bars for
land variable,
#3. need to add labels for land variable in fill= factor(land, levels =c(0,1) ,
labels= c("No","Yes"))
#4. in geom_col we need to add position = "dodge" to add side by side bars for
land variable
#5.. dodge is used to place the bars next to each other instead of stacking them
on top of each other.
#6.  add fill= factor(land) in labs to add legends for land variable
hh|> filter(!is.na(edu_cat), !is.na(land)) |>
      count(edu_cat,land)   |>   mutate(prop   =   n   /   sum(n),   top_lab=
paste0(round(prop*100,1), "%")) |>
  ggplot(aes(x= factor(edu_cat, levels= c(1,2,3),
                   labels = c("Primary", "Secondary", "Higher")),y = prop,fill=
factor(land, levels =c(0,1) , labels= c("No","Yes")))) +
  geom_col(position = "dodge")+ scale_y_continuous(labels= percent)+
  labs(
    x = "Education Categories",
    y = "Percentage",
    title = "Distribution of Education Categories",
    fill = "Land"
  )+
  theme_minimal()+
  theme(
    plot.title = element_text(hjust = 0.5))+
  theme(legend.position = "right")
```

```
Error:
! object 'hh' not found
```

Counts, colours, percentages on the bars and side-by-side bars for a second variable.

Frequency and contingency tables

```
###### FREQUENCY TABLE #######
library(ivo.table)
library(flextable)

## CONTINGENCY TABLE
hh |>
  dplyr::filter(!is.na(edu_cat)) |>
  dplyr::mutate(edu_cat = as.numeric(edu_cat)) |>
  dplyr::select(edu_cat) |>
```

```
  ivo.table::ivo_table(
    color = "black",
    font = "Times New Roman",
    percent_by = "row"
  ) |>
  flextable::save_as_docx(
    path = "Results/frequency_table.docx"
  )
```

```
Error:
! object 'hh' not found
```

```
## install package janitor to clean the frequency tables
library(janitor)
hh|> mutate(across(c(edu_cat, land), as.factor)) |>
  tabyl(edu_cat, land) |>
  adorn_percentages("row") |>
  adorn_pct_formatting() |>
  janitor::adorn_ns() |>
  flextable::save_as_docx(
    path = "Results/freq_1.docx"
  )
```

```
Error:
! object 'hh' not found
```

```
# saving it
hh |>
  dplyr::mutate(across(c(edu_cat, land), haven::as_factor)) |>
  janitor::tabyl(edu_cat, land) |>
  janitor::adorn_percentages("row") |>
  janitor::adorn_pct_formatting() |>
  janitor::adorn_ns() |>
  flextable::flextable() |>
  flextable::save_as_docx(
    path = "Results/freq_1.docx"
  )
```

```
Error:
! object 'hh' not found
```

Cross tables with `janitor`, saved to Word with `flextable`.

Pie chart

```
## simple pie chart
library(ggplot2)
hh|> filter(!is.na(edu_cat)) |>
  count(edu_cat) |>
  mutate(prop = n / sum(n), top_lab= paste0(round(prop*100,1), "%")) |>
  ggplot(aes(x= "", y = prop, fill= factor(edu_cat))) +
  geom_col() + coord_polar(theta = "y") + theme_void()
```

```
Error:
! object 'hh' not found
```

```
## final pie chart with labels and legends
hh|> filter(!is.na(edu_cat)) |>
  count(edu_cat) |>
  mutate(prop = n / sum(n), top_lab= paste0(round(prop*100,1), "%")) |>
  ggplot(aes(x= "", y = prop, fill= factor(edu_cat, levels= c(1,2,3), labels =
c("Primary", "Secondary", "Higher")))) +
  geom_col() + coord_polar(theta = "y") + theme_void() + # coordinate polar is
used to convert bar chart to pie chart
  labs(fill = "Education Categories") +
  geom_text(aes(label = top_lab), position = position_stack(vjust = 0.5))
```

```
Error:
! object 'hh' not found
```

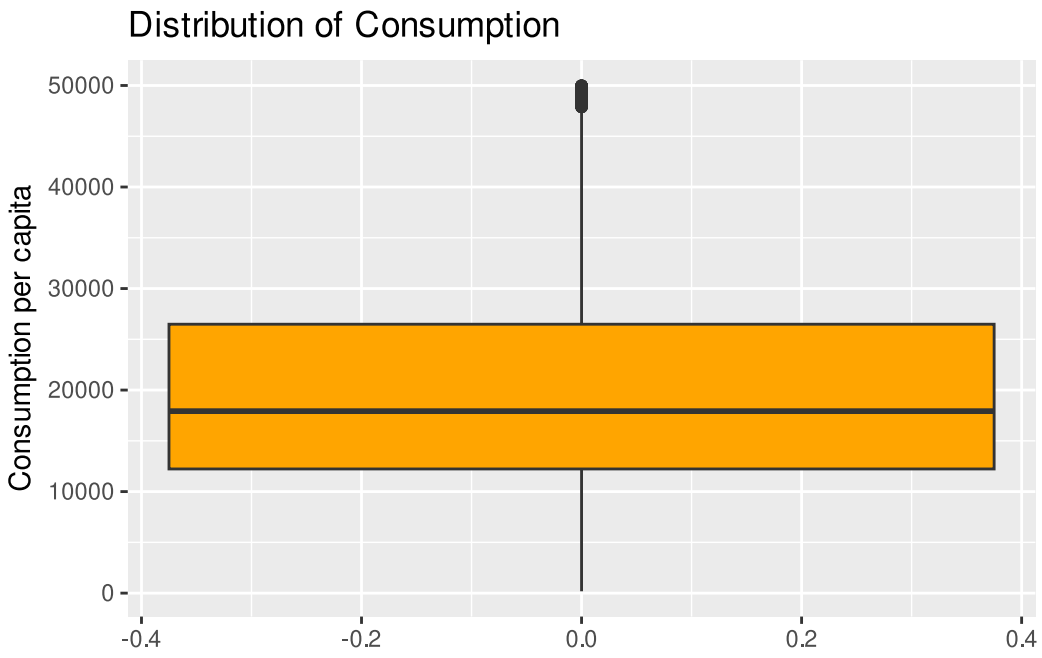
A bar chart turned into a pie with `coord_polar()`, with percentage labels.

Box plot, histogram and density

```
### now th continuous variable
# box plots
ggplot(hh, aes(y = education))+ # to check the distribution of education variable
by education categories
  geom_boxplot(fill= "orange")+ # to create box plot for education variable
  labs(
    y = "Years of Education",
    title = "Distribution of Education"
  )
```

```
Error:
! object 'hh' not found
```

```
##3 using raw data
HH_DATA |>
  dplyr::filter(!is.na(COPC)) |> filter(COPC < 50000) |> # to remove outliers
from the data
  ggplot2::ggplot(ggplot2::aes(y = COPC)) +
  ggplot2::geom_boxplot(fill = "orange") +
  ggplot2::labs(
    y = "Consumption per capita",
    title = "Distribution of Consumption"
  )
```




```
## IN BOX PLOT Q1 , Q2 Q3 are the values of the variable at 25th, 50th and
75th percentiles respectively. The interquartile range (IQR) is the difference
between Q3 and Q1. The whiskers of the box plot extend to the smallest and
largest values within 1.5 times the IQR from the lower and upper quartiles,
respectively. Any data points outside this range are considered outliers and are
plotted as individual points on the box plot.
## raw data ended
```

```
# HISTOGRAM IS THE BAR POT FOR THE CONTINUOUS VARIABLE
ggplot(hh, aes(education, fill="red")) + geom_histogram(bins=8) +
  labs(title="hist_1", x="Education Years", y="Count") +
  scale_x_continuous(breaks=c(0,4,8,12,16)) + labs(fill = "Educ") +
  theme(legend.position = "")
```

```
Error:
! object 'hh' not found
```

```
# FREQUENCY POLYGON
ggplot(hh, aes(x=education, color="blue")) + geom_freqpoly(bins=8) +
  labs(title="FRequency", x="Education Years", y="Count") +
  scale_x_continuous(breaks=c(0,4,8,12,16)) + labs(color = "Educ") +
  theme(legend.position = "")
```

```
Error:
! object 'hh' not found
```

```
# DENSITY PLOT
ggplot(hh, aes(x=education, fill="red")) + geom_density(bw = 0.4) + # bw means
how smooth it will be
  labs(title="Density", x="Education Years", y="Proportion") +
  scale_x_continuous(breaks=c(0,4,8,12,16)) + labs(color = "Educ") +
  theme(legend.position = "")
```

```
Error:
! object 'hh' not found
```

```
# combaining continuous and categorical variable in one graph
# density plot by groups

ggplot(hh, aes(x=education, fill= factor(edu_cat, levels= c(1,2,3), labels =
c("Primary", "Secondary", "Higher")))) +
  geom_density(bw = 0.4, alpha = 0.5) + # alpha is used to adjust the transparency
of the density plot
     labs(title="Density by Education Categories", x="Education Years",
y="Proportion", fill = "Education Categories") +
  scale_x_continuous(breaks=c(0,4,8,12,16)) +
  theme(legend.position = "right")
```

```
Error:
! object 'hh' not found
```

```
# now fill for land variable
ggplot(hh, aes(x=education, fill= factor(land, levels =c(0,1) , labels=
c("No","Yes")))) +
  geom_density(bw = 0.4) +
  labs(title="Density by Land Ownership", x="Education Years", y="Proportion",
fill = "Land Ownership") +
  scale_x_continuous(breaks=c(0,4,8,12,16)) +
  theme(legend.position = "right")
```

```
Error:
! object 'hh' not found
```

```
# replace fill with color to see two lines for two groups
ggplot(hh, aes(x = education,color = factor(land, levels = c(0,1), labels =
c("No","Yes")))) +
  geom_density(bw = 0.4) +
   labs(title = "Density by Land Ownership", x = "Education Years", y =
"Proportion", color = "Land Ownership") +
  scale_x_continuous(breaks = c(0,4,8,12,16)) +
  theme(legend.position = "right")
```

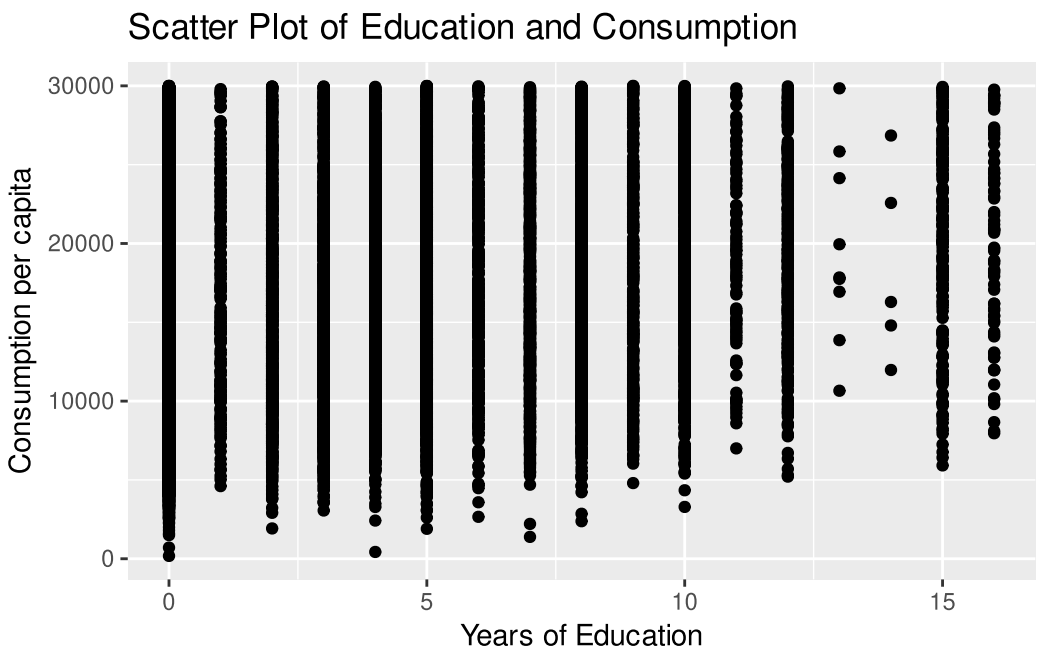
```
Error:
! object 'hh' not found
```

Plots for a continuous variable, and density by group.

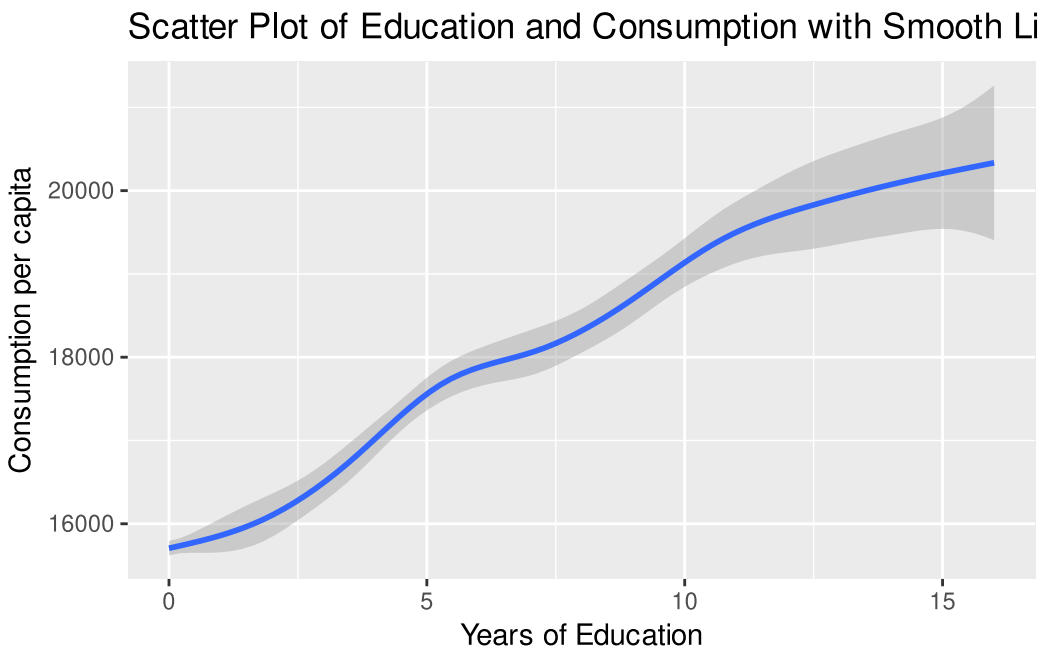
Scatter plots

```
# scatter plots
# plots all the points from the data, we will be the corelation between two
variables, we need to remove NA values and outliers from the data to see the
clear relationship between two variables
HH_DATA |>
  dplyr::filter(!is.na(COPC), !is.na(ID18C)) |> filter(COPC < 30000) |>
  ggplot2::ggplot(ggplot2::aes(x = ID18C, y = COPC)) +
  ggplot2::geom_point() +
  ggplot2::labs(
    x = "Years of Education",
```

```
    y = "Consumption per capita",
    title = "Scatter Plot of Education and Consumption"
  )
```

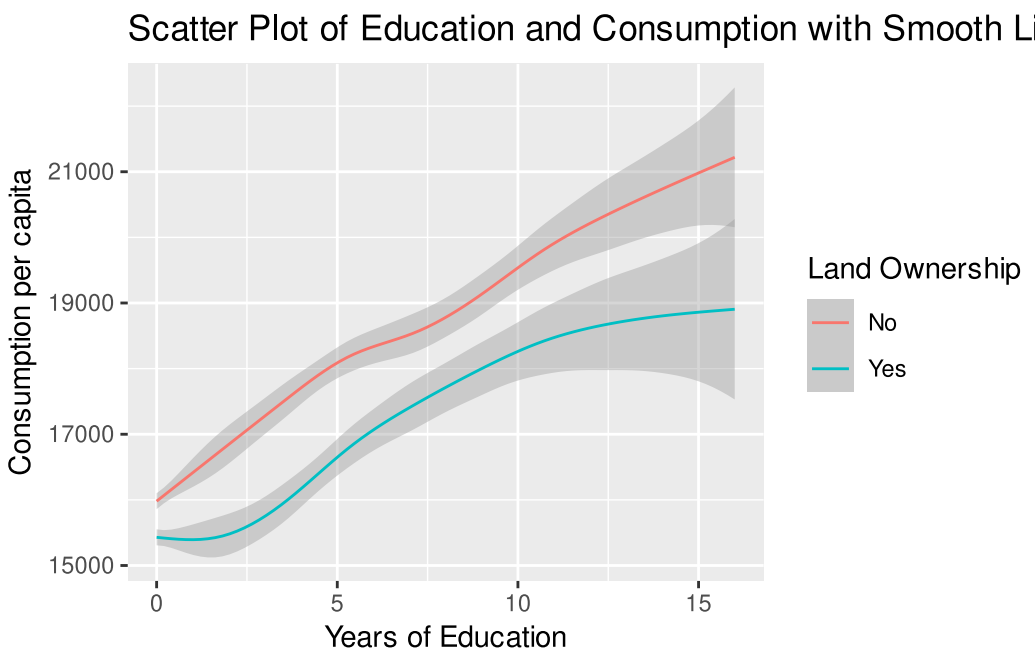



```
# smooth lines
HH_DATA |>
  dplyr::filter(!is.na(COPC), !is.na(ID18C)) |> filter(COPC < 30000) |>
  ggplot2::ggplot(ggplot2::aes(x = ID18C, y = COPC)) +
  ggplot2::geom_smooth(se = T) +
  ggplot2::labs(
    x = "Years of Education",
    y = "Consumption per capita",
    title = "Scatter Plot of Education and Consumption with Smooth Line"
  )
```




```
# se is used to add confidence interval around the smooth line, if se is false
then there will be no confidence interval around the smooth line. The confidence
interval gives us an idea of the uncertainty around the smooth line, it shows us
the range of values within which we can be confident that the true relationship
between the two variables lies. If the confidence interval is narrow, it means
that we can be more confident about the relationship between the two variables,
if it is wide, it means that we are less confident about the relationship between
the two variables.

# by groups
HH_DATA |>
 dplyr::filter(!is.na(COPC), !is.na(ID18C), !is.na(FM1)) |> filter(COPC < 30000)
|>
  ggplot2::ggplot(ggplot2::aes(x = ID18C, y = COPC, color = factor(FM1, levels
= c(0,1), labels = c("No","Yes")))) +
  ggplot2::geom_smooth(se = T, linewidth = 0.5) +
  ggplot2::labs(
    x = "Years of Education",
    y = "Consumption per capita",
     title = "Scatter Plot of Education and Consumption with Smooth Line by
Land Ownership",
    color = "Land Ownership"
  ) +
  theme(legend.position = "right")
```




```
# THESE ARE THE CORELATIONS , BETWEEN , EDUCATION AND CONSUMPTION, AND HOW LAND OWNERSHIP AFFECTS THE RELATIONSHIP BETWEEN EDUCATION AND CONSUMPTION. THE SMOOTH LINE SHOWS US THE TREND OF THE RELATIONSHIP BETWEEN EDUCATION AND CONSUMPTION, IF THE LINE IS UPWARD SLOPING THEN IT MEANS THAT THERE IS A POSITIVE RELATIONSHIP BETWEEN EDUCATION AND CONSUMPTION, IF IT IS DOWNWARD SLOPING THEN IT MEANS THAT THERE IS A NEGATIVE RELATIONSHIP BETWEEN EDUCATION AND CONSUMPTION. THE COLOR OF THE LINE SHOWS US THE DIFFERENCE IN THE RELATIONSHIP BETWEEN EDUCATION AND CONSUMPTION FOR LAND OWNERS AND NON-LAND OWNERS. IF THE LINES ARE SEPARATE THEN IT MEANS THAT THERE IS A DIFFERENCE IN THE RELATIONSHIP BETWEEN EDUCATION AND CONSUMPTION FOR LAND OWNERS AND NON-LAND OWNERS, IF THEY ARE OVERLAPPING THEN IT MEANS THAT THERE IS NO DIFFERENCE IN THE RELATIONSHIP BETWEEN EDUCATION AND CONSUMPTION FOR LAND OWNERS AND NON-LAND OWNERS.
#  THE CORELATION IS POSITIVE FOR BOTH LAND OWNERS AND NON-LAND OWNERS, BUT THE CORELATION IS STRONGER FOR LAND OWNERS THAN NON-LAND OWNERS, BECAUSE THE SMOOTH LINE FOR LAND OWNERS IS MORE UPWARD SLOPING THAN THE SMOOTH LINE FOR NON-LAND OWNERS. THIS MEANS THAT EDUCATION HAS A STRONGER IMPACT ON CONSUMPTION FOR LAND OWNERS THAN NON-LAND OWNERS.
```

Education against consumption, with a smooth line, also by land ownership.